

P.G. open elective

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**SS(Sem-III)—
Phy (9)**

2019(CBCS)

Time : 3 hours

Full Marks : 70

Pass Marks : 32

*Candidates are required to give their answers in
their own words as far as practicable.*

The figures in the margin indicate full marks.

*Answer any **five** questions in which
Q. No. 1 is compulsory.*

1. Answer any **four** of the following questions :

$3\frac{1}{2} \times 4 = 14$

- (a) What is intrinsic semiconductor ?
- (b) Explain atomic packing fraction.
- (c) Explain addition laws in Boolean algebra ?
- (d) Convert 23 into binary number.
- (e) Convert 1101.01 into decimal number.
- (f) Explain principle of LASER action

2. Draw characteristics curve of a Zener diode.
Hence, explain its use as a voltage regulator.

10+4 = 14

3. Draw characteristic of N-P-N transistor in CE mode. Explain how it can be used as an amplifier. 14
4. What are N-channel and P-channel FET ? Draw output characteristics of N-channel FET. 14
5. Explain OR, AND and NOT Gates with the help of truth table. 14
6. Discuss construction and working principle of He-Ne laser. 14
7. What do you mean by nano-particle and quantum dots ? Find out density of states of a quantum wire. 7+7 = 14
8. What do you mean by X-ray diffraction by crystals ? Obtain Bragg's law. Mention its significance. 14

9. Write short notes on any **two** of the following :

7×2 = 14

- (a) Miller indices
- (b) Symmetry operation in crystal
- (c) Characteristics of PN junction
- (d) RUBY laser

